A Floor-Saving Quadratic-over-Logarithmic Lower Bound for Representation Tops in Erdős Problem #1194

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Abstract

Erdős Problem #1194 asks how fast the representation top must grow when $A \subseteq \mathbb{N} = \{1, 2, 3, \ldots\}$ is a perfect difference set in the infinite-integer sense used here, meaning that every integer $n \geq 1$ has a unique ordered representation

$$n = u(n) - v(n), \qquad u(n), v(n) \in A, \qquad u(n) > v(n).$$

Thus $u(n)$ is the larger element in the representation of n, corresponding to the top entry in the original problem statement, not the nth element of A. The main result proves an explicit quadratic-over-logarithmic lower bound in this setting. Let

$$\lambda = \frac{1}{2\log 2}, \qquad B_1 = \frac{1}{2} + \gamma - \log(\log 2),$$

where γ is the Euler–Mascheroni constant. All logarithms in the displayed formulae are natural. For every fixed $B > B_1$, there are arbitrarily large n for which the denominator below is positive and

$$u(n) > \lambda \frac{n^2}{\log n - \log\log n + B}.$$

Equivalently, for this fixed B, the eventual bound $u(n) \leq \lambda n^2/(\log n - \log\log n + B)$ cannot hold. Consequently, for every fixed real C,

$$\limsup_{n\to\infty} \frac{u(n)(\log n - \log\log n + C)}{n^2} \geq \frac{1}{2\log 2}.$$

A companion Lean proof is available at https://github.com/lechmazur/erdos_1194/. No pointwise infinitely-often inequality at $B = B_1$ is asserted.

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All logarithms are natural. Let

$$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

Let

$$\lambda := \frac{1}{2\log 2}.$$

Let γ denote the Euler–Mascheroni constant,

$$\gamma = \lim_{K\to\infty} \left(\sum_{m=1}^{K} \frac{1}{m} - \log K \right),$$

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and put

$$B_1 := \frac{1}{2} + \gamma - \log(\log 2).$$

Numerically,

$$\lambda = 0.721347520444\ldots, \qquad B_1 = 1.443728585483\ldots.$$

Since $0 < \log 2 < 1$, we have $\log(\log 2) < 0$; in particular, $B_1 > 0$.

Asymptotic notation. Throughout, $F \ll_{\mathcal{P}} G$ means $|F| \le C_{\mathcal{P}} G$ on the stated range, $F \gg_{\mathcal{P}} G$ means $G \ll_{\mathcal{P}} F$, and $F \asymp_{\mathcal{P}} G$ means both estimates hold. The same subscript convention is used for O-terms.

Context and status. The hypothesis that every $n \ge 1$ has a unique ordered representation $n = a - b$ with $a, b \in A$ and $a > b$ appears in Erdős's 1980 problem survey [4, p. 100]; it is also catalogued as Erdős Problem #1194 [2]. Here, sets satisfying this hypothesis are called perfect difference sets in this infinite-integer setting; this terminology is local to the present problem and is not intended to invoke the finite cyclic perfect-difference-set convention. Recent work on the finite Sidon-extension conjecture, including counterexamples to that finite conjecture, is logically independent of the infinite representation-top growth function studied here [1]. Throughout, write the unique representation of n as

$$n = u(n) - v(n), \qquad u(n), v(n) \in A, \qquad u(n) > v(n).$$

Then $u(n)$ is the representation top, corresponding to the larger element in the original problem statement, not the nth element of A.

The catalogue page records a greedy construction due to Lev, with growth $u(n) \ll n^3$ in this representation-top notation [6, 2]. It also records the standard Sidon-density argument, based on Erdős's upper bound for infinite Sidon sequences as presented, for example, by Halberstam and Roth, which yields $u(n) \gg n \log n$ infinitely often [5, 2]. The discussion page also records subsequent progress around this problem: Liam Price linked a GPT-5.4 Pro note, and Thomas Bloom distilled from the ensuing discussion an argument giving $a_n \gg n^{3/2}$ infinitely often. Bloom then reported a GPT Pro improvement to $a_n \gg n^{2-o(1)}$ infinitely often, and suggested an optimized $n^2/f(n)$-type form under the convergence condition $\sum_n 1/(nf(n)) < \infty$ [2]. The present draft gives a separate fully written logarithmic lower bound with explicit constants and a companion Lean formalization.

The perfect-difference hypothesis implies the Sidon property that all unordered pair sums $a + b$, with $a \le b$ and $a, b \in A$, are distinct. Indeed, suppose $a \le b$, $c \le d$, and $a + b = c + d$. If $b = d$, then $a = c$. Otherwise, after swapping the two pairs if necessary, $b > d$, and then

$$b - d = c - a > 0.$$

The two ordered pairs cannot coincide: if $(b, d) = (c, a)$, then $b = c$ and $d = a$, so $b > d$ gives $c > a$, while $c \le d = a$ gives $c \le a$. This gives two distinct ordered representations, (b, d) and (c, a), of the same positive difference, contradicting uniqueness. If all representation tops up to difference n were too small, the first n unique difference pairs would force too many elements of A inside a bounded interval, contradicting the classical infinite-Sidon counting bound at that scale. Cilleruelo and Nathanson constructed dense perfect difference sets from dense Sidon sets, showing that the existence theory for perfect difference sets is linked to the density theory of Sidon sets [3].

Scope. Existence is known from the cited constructions. The theorem is universal over all sets satisfying the hypothesis; it does not use or reprove any construction of such sets. The proof starts with an arbitrary set A satisfying the perfect-difference hypothesis and derives the stated lower bound for its representation tops. This draft does not determine the optimal growth of $u(n)$.

Theorem 1. *Let $A \subseteq \mathbb{N}$ be such that, for every $n \geq 1$, there is exactly one ordered pair $(a, b) \in A^2$ with $a > b$ and $a - b = n$. Write this pair as $(u(n), v(n))$, so $n = u(n) - v(n)$ and $u(n) > v(n)$. For every fixed $B > B_1$, there are arbitrarily large integers n such that*

$$u(n) > \lambda \frac{n^2}{\log n - \log\log n + B}.$$

No uniformity as $B \downarrow B_1$ is asserted. The displayed denominator is positive for all sufficiently large n, so there is no large-n sign ambiguity.

Equivalently, for this fixed B, the eventual non-strict bound $u(n) \leq \lambda n^2/(\log n - \log\log n + B)$ cannot hold.

As a limsup consequence, for every fixed real C,

$$\limsup_{n\to\infty} \frac{u(n)(\log n - \log\log n + C)}{n^2} \geq \frac{1}{2\log 2}.$$

Taking $C = B_1$ gives the endpoint limsup version. Since $\log n/(\log n - \log\log n) \to 1$, the conclusion with $C = 0$ also gives

$$\limsup_{n\to\infty} \frac{u(n)\log n}{n^2} \geq \lambda.$$

Remark. Theorem 1 does not determine the optimal growth of $u(n)$. It is a universal lower bound for representation tops, not a construction and not a matching upper bound. The proof below does not establish the pointwise infinitely-often inequality at the endpoint $B = B_1$. The endpoint statement is obtained only by applying the proved result with any fixed $B^* > B_1$ and then taking a limsup along the resulting unbounded sequence; the proof never substitutes $B = B_1$ into the contradiction argument. More generally, any fixed additive constant in the limsup denominator gives the same leading constant. No optimality of B_1 as a sharp pointwise threshold is claimed.

Proof roadmap. The proof argues by contradiction. The assumed upper bound gives a spacing rule for elements of A, which yields a continuous majorant for the counting identity by tops. The integer floor in that identity produces a fixed normalized saving, which contradicts the continuous asymptotic when $B > B_1$. More precisely, the continuous second-order term is reduced by the floor-saving term $2\lambda(1-\gamma)X/\log X$, where $\{q\}$ denotes the fractional part of q; the integral $1-\gamma = \int_1^\infty \{q\}q^{-2}\,dq > 0$ appears after the later change of variables $q = X/\psi(t)$. In compressed form, the dependency chain is

$$\text{spacing} \Rightarrow \text{counting majorant} \Rightarrow J_B(X) - E(X) \Rightarrow \text{a negative } X/\log X \text{ coefficient.}$$

Constant bookkeeping. The final comparison uses

$$\lambda = \frac{1}{2\log 2}, \qquad B_1 = \frac{1}{2} + \gamma - \log(\log 2),$$

and the two second-order coefficients

$$C_{\text{cont}}(B) := \lambda\big(3 - 2B - 2\log(\log 2)\big), \qquad C_{\text{floor}} := 2\lambda(1-\gamma).$$

At the endpoint,

$$C_{\text{cont}}(B_1) = C_{\text{floor}} = 0.609948863612\ldots.$$

The continuous coefficient minus the floor-saving coefficient is

$$C_{\text{cont}}(B) - C_{\text{floor}} = \lambda\big(3 - 2B - 2\log(\log 2)\big) - 2\lambda(1-\gamma) = -2\lambda(B - B_1).$$

The main auxiliary objects below are the eventual majorant Φ, its inverse ψ, the nonincreasing function $f = 1/\psi$ on the inverse tail, the top-window integral G_X, the continuous majorant $J_B(X)$, and the floor saving $E(X)$.

Proof. Use the notation $u(n), v(n)$ from the theorem. Fix

$$B > B_1.$$

All thresholds and implicit constants below are attached to the fixed set A, the fixed parameter B, and the contradiction data introduced below; no uniformity over all admissible sets A, or as $B \downarrow B_1$, is asserted. Recall that

$$\lambda = \frac{1}{2\log 2}.$$

Since $B > B_1$ and $B_1 > 0$, we have $B > 0$. Put

$$H(w) := w - \log w + B \qquad (w > 0).$$

For $w > 0$, the elementary inequality $w - \log w \geq 1$, together with $B > B_1 > 0$, gives

$$H(w) > 0.$$

Also

$$H'(w) = 1 - \frac{1}{w},$$

and

$$2H(w) - H'(w) = 2(w - \log w) + 2B - 1 + \frac{1}{w} \geq 1 + 2B + \frac{1}{w} > 0.$$

Thus, for every $x > 1$, define

$$\Phi(x) := \lambda \frac{x^2}{\log x - \log\log x + B} = \lambda \frac{x^2}{H(\log x)}.$$

Then

$$\Phi(e^w) = \lambda \frac{e^{2w}}{H(w)} \qquad (w > 0),$$

and

$$\frac{d}{dw}\Phi(e^w) = \lambda e^{2w} \frac{2H(w) - H'(w)}{H(w)^2} > 0.$$

Also, as $x \to \infty$,

$$\Phi(x) \sim \lambda \frac{x^2}{\log x} \to \infty.$$

Assume, for contradiction, that

$$u(n) \leq \Phi(n) \tag{1}$$

for all sufficiently large n.

Choose and fix an integer $N_0 \geq 3$ large enough that

$$u(n) \leq \Phi(n) \qquad (n \geq N_0),$$

and that

$$\frac{1}{2}\log x \leq H(\log x) \leq 2\log x, \qquad \log x \leq 2H(\log x) - H'(\log x) \leq 4\log x \qquad (x \geq N_0).$$

Such an N_0 exists because the assumed bound holds eventually and

$$H(\log x) \sim \log x, \qquad 2H(\log x) - H'(\log x) \sim 2\log x \qquad (x \to \infty).$$

The displayed comparisons are used only for tail estimates; positivity of $H(\log x)$ and $2H(\log x) - H'(\log x)$ already holds for every $x > 1$. Since

$$\Phi'(x) = \lambda x \frac{2H(\log x) - H'(\log x)}{H(\log x)^2},$$

we have $\Phi'(x) > 0$ for every $x \geq N_0$. Let

$$T_* := \Phi(N_0).$$

Since $\lambda > 0$, $N_0 > 0$, and $H(\log N_0) > 0$, the definition $T_* = \lambda N_0^2/H(\log N_0)$ gives $T_* > 0$. Since Φ is continuous, strictly increasing on $[N_0, \infty)$, and tends to infinity, it maps $[N_0, \infty)$ bijectively onto $[T_*, \infty)$. Let ψ be the inverse of Φ on this interval. Since Φ is C^1 and $\Phi' > 0$ on $[N_0, \infty)$, the inverse ψ is C^1 on (T_*, ∞). All later uses of ψ occur at arguments at least T_*, after the relevant lower thresholds have been imposed. Thus whenever ψ appears below, it denotes this branch only; inversions of Φ are never taken outside $[N_0, \infty)$.

Define

$$f(t) := \frac{1}{\psi(t)} \qquad (t \geq T_*).$$

Extend f to $[0, T_*)$ by

$$f(t) := f(T_*) \qquad (0 \leq t < T_*).$$

Then f is positive, nonincreasing, locally bounded, and measurable on $[0, \infty)$. Indeed, ψ is increasing on $[T_*, \infty)$, so $1/\psi$ is nonincreasing on $[T_*, \infty)$. Also $\psi(T_*) = N_0$, and hence $f(t) \leq f(T_*)$ for $t \geq T_*$. Therefore the constant extension to $[0, T_*)$ preserves monotonicity across the joining point. Moreover, f is C^1 on the smooth tail (T_*, ∞). All integrals below are Lebesgue integrals.

The artificial extension on $[0, T_*)$ is only a bookkeeping device. Only the values $t \geq T_*$ are tied to the inverse branch ψ; values below T_* are used solely inside integrals. Since T_* is fixed and f is bounded there, every contribution from this interval is $O_{A,B}(1)$, hence negligible on the scales $\sqrt{X/\log X}$ and $X/\log X$ used later. It is also negligible in the normalized L^1 rescaling used below, where its contribution is $O_{A,B}(\sqrt{Y/X})$.

Lemma 1 (Uniform inverse-branch convention on compact q-ranges)**.** *For every fixed $Q \geq 1$, all sufficiently large X satisfy $X/Q \geq N_0$, and then*

$$\psi(\Phi(X/q)) = X/q \qquad (1 \leq q \leq Q).$$

Consequently, after increasing the lower threshold in X when necessary, every later occurrence of ψ is evaluated on the inverse branch from $[T_*, \infty)$ to $[N_0, \infty)$ fixed above.

Proof. For fixed $Q \geq 1$, the inequality $X/Q \geq N_0$ holds for all sufficiently large X. Then $X/q \geq X/Q \geq N_0$ for every $1 \leq q \leq Q$, and the identity follows from the definition of the inverse branch. The displayed choices used later, including $M = \Phi(X)$, $\Phi(X/Q)$, and $\Phi(X/q)$ for fixed Q and $1 \leq q \leq Q$, are covered by this compact-range observation; separately, $X \geq T_*$ for all sufficiently large X. □

1. Spacing

Write

$$A = \{x_1 < x_2 < x_3 < \cdots\}.$$

This enumeration is infinite because every positive integer occurs as a difference of two elements of A. Hence A is unbounded.

Since $N_0 \geq 3$, the finite maximum below is over a nonempty set. Choose an integer

$$Z_0 \in \mathbb{N}, \qquad Z_0 > \max\left(T_*, \max_{1 \leq d < N_0} u(d)\right).$$

The integrality of Z_0 is used only when bounding integer tops in the short interval $(X, X + Z_0]$.

Threshold and uniformity convention. Lower thresholds may be enlarged finitely many times after the fixed data A, B, N_0, T_*, Z_0 and any fixed truncation parameter have been chosen. Thus all later occurrences of $\log\log$, Φ, ψ, and denominators of the form $\log z - \log\log z + C$ are on ranges where they are defined and have the asserted sign.

Unless explicitly stated otherwise, all O- and o-terms are taken after the contradiction data have been fixed. Constants depending on A, B, N_0, T_*, Z_0 are written as $O_{A,B}$; lower thresholds may also depend on previously fixed auxiliary parameters. If a truncation parameter such as U or Q appears, it is fixed before $X \to \infty$ and is sent to its limiting value only after that limit. Any tail estimate later used after U or Q is sent to infinity states the needed independence of the displayed implicit constant; no truncation parameter varies with X. Whenever $Y = \log X$ is used below, the scale separations $X/Y \to \infty$ and $X/Y^2 \to \infty$ are also used only on this eventual range.

The notation convention stated above remains in force. Extra subscripts record extra fixed data on which the implicit constant may depend.

Spacing and gap-integral domination

Lemma 2 (Spacing and gap-integral domination)**.** *If $y < z$ are elements of A and $z > Z_0$, then*

$$z - y \geq \psi(z). \tag{2}$$

Consequently,

$$\int_y^z f(s)\, ds \geq 1. \tag{3}$$

Proof. Put

$$d := z - y.$$

Then d is a positive integer. The ordered pair (z, y) represents d. By uniqueness of the representing pair for d,

$$u(d) = z.$$

If $d < N_0$, then

$$z = u(d) \leq \max_{1 \leq m < N_0} u(m) < Z_0,$$

contradicting $z > Z_0$. Hence $d \geq N_0$. By the assumed upper bound (1),

$$z = u(d) \leq \Phi(d).$$

Since $d \geq N_0$, $z > Z_0 > T_*$, and Φ is increasing on $[N_0, \infty)$, applying the inverse branch $\psi : [T_*, \infty) \to [N_0, \infty)$ to $z \leq \Phi(d)$ gives

$$d \geq \psi(z).$$

Thus $z - y = d \geq \psi(z)$, proving (2).

Since f is nonincreasing, $f(s) \geq f(z)$ for $y \leq s \leq z$. Also $z > Z_0 > T_*$ gives $f(z) = 1/\psi(z)$. Hence

$$\int_y^z f(s)\,ds \geq (z-y)f(z) = \frac{z-y}{\psi(z)} \geq 1.$$

□

Counting-function majorant

For real $T \geq 0$, define

$$\mathcal{N}_A(T) := |A \cap [1,T]|,$$

where $[1,T]$ is understood to be empty if $0 \leq T < 1$.

Lemma 3 (Counting-function majorant)**.** *For all real $T \geq 0$,*

$$\mathcal{N}_A(T) \leq C_* + \int_0^T f(s)\,ds \tag{4}$$

for some constant $C_* = C(A, B, N_0, T_*, Z_0)$.

Proof. In the argument below one may take $C_* = j_0$ after j_0 has been defined. By unboundedness and the well-ordering of $\mathbb{N}$, there is a first element of A exceeding Z_0; call it x_{j_0}. If $\mathcal{N}_A(T) \leq j_0$, then (4) holds with $C_* \geq j_0$. Otherwise, let $m = \mathcal{N}_A(T) > j_0$. Then $x_m \leq T$, and by (3),

$$m - j_0 \leq \sum_{j=j_0+1}^{m} \int_{x_{j-1}}^{x_j} f(s)\,ds \leq \int_0^T f(s)\,ds.$$

Thus

$$\mathcal{N}_A(T) = m \leq j_0 + \int_0^T f(s)\,ds,$$

which proves (4) with $C_* = j_0$. □

2. Estimates for f

For the tail estimates in this part, let $t > T_*$ and write

$$r = \psi(t).$$

Then

$$t = \Phi(r) = \lambda \frac{r^2}{\log r - \log\log r + B} = \lambda \frac{r^2}{H(\log r)}.$$

As $t \to \infty$, also $r \to \infty$. Since

$$H(\log r) \asymp_B \log r$$

for large r, we have

$$\log t = 2\log r - \log H(\log r) + O_B(1) = 2\log r + O_B(\log\log r).$$

Since $O_B(\log\log r) = o(\log r)$, this first implies

$$\log r \asymp_B \log t.$$

Consequently $\log H(\log r) = O_B(\log\log t)$, because $H(\log r) \asymp_B \log r$ on the tail. Substituting this back into

$$\log t = 2\log r - \log H(\log r) + O_B(1)$$

gives

$$\log r = \frac{1}{2}\log t + O_B(\log\log t).$$

Equivalently,

$$\log r = \frac{1}{2}\log t\left(1 + O_B\left(\frac{\log\log t}{\log t}\right)\right).$$

Therefore

$$\log\log r = \log\log t + \log\left(\frac{1}{2} + O_B\left(\frac{\log\log t}{\log t}\right)\right) = \log\log t + O_B(1).$$

Also,

$$f(t)^2 = \frac{1}{r^2} = \frac{\lambda}{t\,H(\log r)}.$$

Because

$$H(\log r) = \log r - \log\log r + B = \frac{1}{2}\log t + O_B(\log\log t),$$

we have

$$\frac{1}{H(\log r)} = \frac{2}{\log t}\left(1 + O_B\left(\frac{\log\log t}{\log t}\right)\right).$$

Together with $f(t)^2 = \lambda/(tH(\log r))$, and using that the factor in parentheses is $1 + o(1) > 0$ on the tail and $\sqrt{1 + O(\varepsilon)} = 1 + O(\varepsilon)$ for small ε, this gives after taking the positive square root, uniformly for all sufficiently large t,

$$f(t) = \sqrt{\frac{2\lambda}{t\log t}}\left(1 + O_B\left(\frac{\log\log t}{\log t}\right)\right). \tag{5}$$

In particular,

$$f(t) \ll_B \frac{1}{\sqrt{t\log t}} \qquad (t \to \infty). \tag{6}$$

Integral asymptotic for f

Lemma 4 (Integral asymptotic for f)**.** *As $X \to \infty$,*

$$\int_0^X f(t)\,dt = 2\sqrt{\frac{2\lambda X}{\log X}} + o\left(\sqrt{\frac{X}{\log X}}\right). \tag{7}$$

Proof. We first record the elementary estimate

$$\int_{T_1}^V \frac{dt}{\sqrt{t\log t}} \ll \sqrt{\frac{V}{\log V}} \qquad (V \to \infty), \tag{8}$$

for any fixed sufficiently large T_1. For $V \geq e^2$,

$$\frac{d}{dV}\sqrt{\frac{V}{\log V}} = \frac{\log V - 1}{2\sqrt{V}(\log V)^{3/2}} \asymp \frac{1}{\sqrt{V\log V}},$$

so increasing T_1 if necessary and integrating gives (8).

Now put

$$Y := \log X.$$

Since $X/Y^2 \to \infty$, all thresholds below are exceeded for sufficiently large X. The bounded initial interval contributes $O_{A,B}(1)$. By (6), for any fixed sufficiently large T_1,

$$\int_0^{X/Y^2} f(t)\,dt = O_{A,B}(1) + O_B\left(\int_{T_1}^{X/Y^2} \frac{dt}{\sqrt{t\log t}}\right).$$

Using (8), the last integral is

$$\ll_B \sqrt{\frac{X/Y^2}{\log(X/Y^2)}} \ll_B \frac{\sqrt{X}}{Y^{3/2}} = o\left(\sqrt{\frac{X}{Y}}\right).$$

Thus

$$\int_0^{X/Y^2} f(t)\,dt = o\left(\sqrt{\frac{X}{Y}}\right).$$

On the interval $X/Y^2 \le t \le X$, we have

$$\log t = Y + O(\log Y)$$

uniformly. Hence the error term in (5) is uniformly $o(1)$ on this interval; explicitly,

$$\sup_{X/Y^2 \le t \le X} \left| \frac{f(t)}{\sqrt{2\lambda/(tY)}} - 1 \right| \to 0.$$

Therefore

$$f(t) = \sqrt{\frac{2\lambda}{Y}}\, t^{-1/2}(1 + o(1))$$

uniformly for $X/Y^2 \le t \le X$. Also,

$$\int_{X/Y^2}^{X} t^{-1/2}\,dt = 2\sqrt{X}\left(1 - \frac{1}{Y}\right) = 2\sqrt{X} + O\left(\frac{\sqrt{X}}{Y}\right).$$

It follows that

$$\int_{X/Y^2}^{X} f(t)\,dt = 2\sqrt{\frac{2\lambda X}{Y}} + o\left(\sqrt{\frac{X}{Y}}\right),$$

which proves (7). □

Since Z_0 is fixed, the same estimate remains valid with $X + Z_0$ in place of X. Indeed, apply (7) with $X + Z_0$ in place of X; writing $Y = \log X$,

$$\sqrt{\frac{X + Z_0}{\log(X + Z_0)}} \sim \sqrt{\frac{X}{\log X}} = \sqrt{\frac{X}{Y}}.$$

Thus

$$\int_0^{X+Z_0} f(t)\,dt = 2\sqrt{\frac{2\lambda X}{Y}} + o\left(\sqrt{\frac{X}{Y}}\right) = O_{A,B}\left(\sqrt{\frac{X}{Y}}\right).$$

Together with (4), this gives

$$\mathcal{N}_A(X + Z_0) = O_{A,B}\left(\sqrt{\frac{X}{Y}}\right). \tag{9}$$

Finally, on the smooth tail $t > T_*$,

$$|f'(t)| \ll_B \frac{f(t)}{t}. \tag{10}$$

Indeed, if $t = \Phi(r)$, then $r = \psi(t) \geq N_0$ and $f(t) = 1/r$, so

$$f'(t) = -\frac{1}{r^2\Phi'(r)}.$$

Now

$$\Phi'(r) = \lambda r \frac{2H(\log r) - H'(\log r)}{H(\log r)^2}.$$

Therefore

$$|f'(t)| = \frac{H(\log r)^2}{\lambda r^3\big(2H(\log r) - H'(\log r)\big)}.$$

Since

$$H(w) \sim w, \qquad 2H(w) - H'(w) \sim 2w,$$

and since the displayed bounds imposed when choosing N_0 apply with $x = r$ for every $r \geq N_0$, these comparisons hold with uniform upper and lower constants. Hence

$$\Phi'(r) \asymp_B \frac{r}{H(\log r)},$$

and the exact identity above gives

$$|f'(t)| \asymp_B \frac{H(\log r)}{r^3}.$$

On the other hand,

$$t \asymp_B \frac{r^2}{H(\log r)}, \qquad f(t) = \frac{1}{r}.$$

Therefore

$$|f'(t)| \ll_B \frac{f(t)}{t}.$$

Lemma 5 (Local comparability and Lipschitz control)**.** *For every fixed $c \in (0,1)$, there is a threshold $T(c) > T_*/c$ such that, whenever $t \geq T(c)$ and $ct \leq s \leq t$,*

$$f(s) \asymp_{B,c} f(t), \tag{11}$$

and

$$|f(s) - f(t)| \ll_{B,c} \frac{f(t)}{t}\,|t - s|. \tag{12}$$

For fixed c, the implicit constants here depend only on B and c, not on the later truncation parameters U or Q.

Proof. If $ct \le s \le t$, then

$$\log s = \log t + O_c(1),$$

so $s^{-1/2} \asymp_c t^{-1/2}$ and $\log s \asymp_c \log t$. Applying (5) to both s and t, after increasing $T(c)$ if necessary, gives (11). For $t \ge T(c)$, the full interval $[s,t]$ lies in the smooth tail (T_*, ∞). If $\xi \in [s,t]$, then $ct \le \xi \le t$, so (11), applied with ξ in place of s, gives

$$f(\xi) \asymp_{B,c} f(t).$$

By (10),

$$|f'(\xi)| \ll_B \frac{f(\xi)}{\xi} \ll_{B,c} \frac{f(t)}{t}.$$

The mean-value theorem now gives (12). □

3. Counting by tops

This section records the exact top-counting identity used in the contradiction.

From now on, X tends to infinity through positive integers in counting identities. The same symbol X is also used as a real analytic parameter, for example as an endpoint of integration or as an argument of Φ and ψ.

Let $X \to \infty$ through positive integers and put

$$Y := \log X, \qquad M := \Phi(X).$$

The quantity M need not be an integer. Sums over $t \in A$ with $t \le M$ are sums over integer elements of A bounded by the real number M. The analytic estimates are applied to the same integer values of X, with X also used as a real parameter inside integrals. Throughout the remaining proof, X is assumed large enough for every analytic interval whose nonemptiness is required to be nonempty and for all previously fixed thresholds to be exceeded. Discrete sets such as $A \cap (X + Z_0, M]$ may of course be empty and are handled separately.

For all sufficiently large integer X, every $1 \le n \le X$ has

$$u(n) \le M.$$

Indeed, for the finitely many $n < N_0$, each $u(n)$ is fixed and $M = \Phi(X) \to \infty$, so this holds once X is large enough. For $n \ge N_0$, the assumed bound (1) gives

$$u(n) \le \Phi(n) \le \Phi(X) = M,$$

because $n \le X$ and Φ is increasing on $[N_0, \infty)$.

Also,

$$M = \Phi(X) \asymp_B \frac{X^2}{Y},$$

so $M > X + Z_0$ for all sufficiently large X, since Z_0 is fixed. Since every difference $1 \le n \le X$ has top at most M, counting differences by their top gives the following exact identity.

Lemma 6 (Top-counting identity)**.** *For all sufficiently large integer X, the exact top-counting identity is*

$$X = \sum_{\substack{t \in A \\ t \le M}} \#\{b \in A : t - X \le b < t\}. \tag{13}$$

Proof. Each $1 \le n \le X$ contributes its unique representing pair

$$n = u(n) - v(n),$$

whose top satisfies $u(n) \le M$, so the map from counted pairs to their differences is surjective onto $\{1, 2, \ldots, X\}$. Conversely, every counted pair (t, b) gives a positive difference

$$t - b \in [1, X].$$

Uniqueness ensures that two counted pairs cannot give the same difference, so this map is injective on the counted pairs. Therefore the count is exactly X. □

No loss has been introduced in (13); all later inequalities enter only after splitting the exact count. We split the right-hand side according as $t \le X$ or $t > X$.

4. Small tops

Since $M > X$, the portion of (13) with $t \le X$ is exactly

$$\binom{\mathcal{N}_A(X)}{2}.$$

Each two-element subset of $A \cap [1, X]$ contributes exactly one ordered pair, with the larger element as top and the smaller element as bottom. Indeed, if $t \le X$, then every $b \in A$ with $b < t$ gives a difference

$$1 \le t - b \le X.$$

Conversely, since $A \subseteq \mathbb{N}$, every such b satisfies $b \ge 1$; and since $t - X \le 0$, the condition $t - X \le b < t$ in (13) is automatic for every $b \in A$ with $b < t$. By (4), there is a fixed constant $C_{\mathrm{sp}} = C_{\mathrm{sp}}(A, B, N_0, T_*, Z_0)$ such that

$$\mathcal{N}_A(X) \le F(X) + C_{\mathrm{sp}}, \qquad F(X) := \int_0^X f(t)\, dt.$$

The artificial extension of f on $[0, T_*)$ changes $F(X)$ only by a fixed $O_{A,B}(1)$ amount relative to the smooth tail. The discarded term in

$$\binom{\mathcal{N}_A(X)}{2} = \frac{1}{2}\mathcal{N}_A(X)^2 - \frac{1}{2}\mathcal{N}_A(X)$$

is harmless at the present scale, because $\mathcal{N}_A(X) \le \mathcal{N}_A(X + Z_0) = O_{A,B}(\sqrt{X/Y})$ by (9), and

$$\sqrt{X/Y} = o(X/Y).$$

Therefore the cruder bound

$$\binom{\mathcal{N}_A(X)}{2} \le \frac{1}{2}\mathcal{N}_A(X)^2 \le \frac{1}{2}(F(X) + C_{\mathrm{sp}})^2$$

is sharp enough for the second-order comparison. Expanding,

$$\frac{1}{2}(F(X) + C_{\mathrm{sp}})^2 = \frac{1}{2}F(X)^2 + O_{A,B}(F(X)) + O_{A,B}(1).$$

By (7),

$$F(X) = 2\sqrt{\frac{2\lambda X}{\log X}} + o\left(\sqrt{\frac{X}{\log X}}\right),$$

and no second-order expansion of F is needed. Hence

$$O_{A,B}(F(X)) + O_{A,B}(1) = o\left(\frac{X}{\log X}\right).$$

Thus

$$\binom{\mathcal{N}_A(X)}{2} \le \frac{1}{2}\left(\int_0^X f(t)\,dt\right)^2 + o\left(\frac{X}{Y}\right). \tag{14}$$

By (7),

$$\frac{1}{2}\left(\int_0^X f(t)\,dt\right)^2 = \frac{4\lambda X}{Y} + o\left(\frac{X}{Y}\right). \tag{15}$$

5. Large tops and the integer floor

The only improvement over the continuous large-top majorant is that the integer count attached to a top t is bounded by $\lfloor G_X(t)\rfloor$, not merely by $G_X(t)$. For $t \ge X$, define

$$G_X(t) := \int_{t-X}^{t} f(s)\,ds.$$

Since f is nonincreasing, G_X is also nonincreasing. Indeed, for $h > 0$,

$$G_X(t+h) = \int_{t-X}^{t} f(s+h)\,ds \le \int_{t-X}^{t} f(s)\,ds = G_X(t).$$

Hence $\lfloor G_X(t)\rfloor$ is also nonincreasing. Since G_X is monotone, it is measurable, and therefore $\lfloor G_X\rfloor$ is measurable. On each compact interval $[X, M]$,

$$0 \le G_X(t) \le \int_0^X f(s)\,ds,$$

because $t - X \ge 0$ and f is nonincreasing. Thus $G_X(t)$ is finite for every $t \in [X, M]$, $\lfloor G_X\rfloor$ is bounded on $[X, M]$, and $f(t)\lfloor G_X(t)\rfloor$ is measurable and integrable there.

Fix $t \in A$ with

$$t > X + Z_0.$$

Let

$$r_t := \#\{b \in A : t - X \le b < t\}.$$

If $r_t = 0$, then $r_t \le \lfloor G_X(t)\rfloor$ is trivial because $G_X(t) \ge 0$. Assume $r_t \ge 1$, and list the contributing bottoms as

$$b_1 < b_2 < \cdots < b_{r_t} < t,$$

with

$$b_i \in A, \qquad t - X \le b_i < t.$$

Because $t > X + Z_0$, each b_i satisfies

$$b_i \ge t - X > Z_0.$$

Set $b_{r_t+1} := t$. Then

$$b_1 < b_2 < \cdots < b_{r_t} < b_{r_t+1} = t$$

has exactly r_t gaps, including the final gap ending at t. For every $1 \le i \le r_t$, the upper endpoint b_{i+1} exceeds Z_0, so (3) applies to the gap $[b_i, b_{i+1}]$. Hence

$$r_t \le \sum_{i=1}^{r_t} \int_{b_i}^{b_{i+1}} f(s)\,ds = \int_{b_1}^{t} f(s)\,ds \le \int_{t-X}^{t} f(s)\,ds = G_X(t).$$

The quantity $G_X(t)$ is nonnegative, and r_t is an integer. Hence the inequality $r_t \le G_X(t)$ implies

$$r_t \le \lfloor G_X(t) \rfloor. \tag{16}$$

Let

$$t_1 < t_2 < \cdots < t_L$$

be the elements of $A \cap (X + Z_0, M]$. If $L = 0$, there is no contribution from this range. Assume $L \ge 1$. For $j \ge 2$, the elements t_{j-1} and t_j are elements of A with $t_{j-1} < t_j$ and

$$t_j > X + Z_0 > Z_0.$$

The spacing estimate (3), which only requires two elements $y < z$ of A with $z > Z_0$, therefore applies to the gap $[t_{j-1}, t_j]$.

The next step converts the discrete sum over large tops into a gap-integral sum, losing only the first large top. Because $\lfloor G_X(t) \rfloor$ is nonincreasing,

$$\int_{t_{j-1}}^{t_j} f(s)\lfloor G_X(s) \rfloor\,ds \ge \lfloor G_X(t_j) \rfloor \int_{t_{j-1}}^{t_j} f(s)\,ds \ge \lfloor G_X(t_j) \rfloor.$$

Therefore, if $L \ge 2$,

$$\sum_{j=2}^{L} \lfloor G_X(t_j) \rfloor \le \int_{t_1}^{t_L} f(t)\lfloor G_X(t) \rfloor\,dt \le \int_{X}^{M} f(t)\lfloor G_X(t) \rfloor\,dt.$$

Here extending the integral to all of $[X, M]$ is legitimate because

$$f(t)\lfloor G_X(t) \rfloor \ge 0.$$

The first top t_1 is the only possible loss in this integral domination. Adding this first-top loss gives

$$\sum_{j=1}^{L} \lfloor G_X(t_j) \rfloor \le \lfloor G_X(t_1) \rfloor + \int_{X}^{M} f(t)\lfloor G_X(t) \rfloor\,dt. \tag{17}$$

For $L = 1$, the same inequality is immediate because the integral term is nonnegative. Thus (17) holds for all $L \ge 1$.

The endpoint term is negligible. Since $t_1 > X$,

$$\lfloor G_X(t_1) \rfloor \le G_X(t_1) = \int_{t_1-X}^{t_1} f(s)\,ds.$$

Here $t_1 - X > 0$. Among intervals of length X in $[0, \infty)$, the integral of a nonincreasing function is maximized by $[0, X]$. Indeed, for $a \geq 0$,

$$\int_a^{a+X} f(s)\, ds = \int_0^X f(u+a)\, du \leq \int_0^X f(u)\, du.$$

Therefore, by (7),

$$G_X(t_1) \leq \int_0^X f(s)\, ds = O_{A,B}\left(\sqrt{\frac{X}{Y}}\right) = o\left(\frac{X}{Y}\right). \tag{18}$$

Consequently, if $L = 0$, the estimate below is immediate because the left side is zero and the integral is nonnegative. If $L \geq 1$, (16), (17), and (18) give the same estimate. Hence, in all cases,

$$\sum_{t \in A \cap (X+Z_0, M]} r_t \leq \int_X^M f(t) \lfloor G_X(t) \rfloor\, dt + o\left(\frac{X}{Y}\right).$$

The integer tops in the short interval $X < t \leq X + Z_0$ also contribute only $o(X/Y)$. Indeed, since $A \subseteq \mathbb{N}$ and X and Z_0 are integers, the interval $(X, X + Z_0] \cap \mathbb{N}$ contains at most Z_0 possible integer tops. This is the only use of the integrality of Z_0. Each such top has at most $\mathcal{N}_A(X + Z_0)$ possible bottoms. By (9), the total contribution from these tops is at most

$$Z_0 \mathcal{N}_A(X + Z_0) = O_{A,B}\left(\sqrt{\frac{X}{Y}}\right) = o\left(\frac{X}{Y}\right).$$

Combining (13), (14), and the preceding large-top bounds, we get

$$X \leq \frac{1}{2}\left(\int_0^X f(t)\, dt\right)^2 + \int_X^M f(t) \lfloor G_X(t) \rfloor\, dt + o\left(\frac{X}{Y}\right). \tag{19}$$

It remains to evaluate the right-hand side sharply.

6. The continuous majorant without the floor

Throughout parts 6 through 8,

$$Y = \log X,$$

with $X \to \infty$ through positive integers. Also $M = \Phi(X)$; as observed above, $M \asymp_B X^2/Y > X$ for all sufficiently large X. Later $t = \Phi(r)$, $r = \psi(t)$, and $q = X/r$, while U and Q are fixed real truncation parameters before the limit $X \to \infty$ is taken. The lower-endpoint expansion feeds both the square term and the large-q floor tail; local Lipschitz control feeds the correction-tail estimate; and the floor-saving main term uses the fixed-Q change of variables together with $G_X(\Phi(X/q)) = q + o(1)$ uniformly on fixed q-ranges. For tail estimates below that are later used after a truncation parameter is sent to infinity, implicit constants are independent of that truncation parameter, while lower X-thresholds may depend on its fixed value. Dependence on a fixed truncation parameter is displayed in subscripts such as $o_{A,B,U}$ or $o_{A,B,Q}$. Define

$$J_B(X) := \frac{1}{2}\left(\int_0^X f(t)\, dt\right)^2 + \int_X^M f(t) G_X(t)\, dt.$$

We prove

$$J_B(X) = X + \frac{\lambda\big(3 - 2B - 2\log(\log 2)\big)X}{Y} + o\left(\frac{X}{Y}\right). \tag{20}$$

The proof of this asymptotic is split into the square term, the correction term on a fixed truncation range, and a correction-tail estimate whose normalized constant is independent of the truncation parameter.

The small-top term is already known from (15):

$$\frac{1}{2}\left(\int_0^X f(t)\,dt\right)^2 = \frac{4\lambda X}{Y} + o\left(\frac{X}{Y}\right). \tag{21}$$

Next,

$$\int_X^M f(t)G_X(t)\,dt = X\int_X^M f(t)^2\,dt + R(X), \tag{22}$$

where

$$R(X) := \int_X^M f(t)\int_{t-X}^t \big(f(s)-f(t)\big)\,ds\,dt.$$

Because f is nonincreasing and $s \le t$ inside the inner integral, $R(X) \ge 0$.

6.1. The square term and constant computation

Lemma 7 (Lower endpoint expansion)**.** *As $X \to \infty$ through positive integers, let*

$$r_0 = r_0(X) := \psi(X), \qquad w_0 = w_0(X) := \log r_0.$$

Then

$$w_0 = \frac{Y}{2} + \frac{1}{2}\log Y - \frac{1}{2}\log(2\lambda) + o(1). \tag{23}$$

Equivalently,

$$r_0 = \psi(X) = \sqrt{\frac{XY}{2\lambda}}\,(1+o(1)).$$

Consequently, for every fixed $Q > 1$,

$$r_0 = \psi(X) = o(X/Q).$$

Proof. For all sufficiently large X, we have $X > T_*$, so $r_0 = \psi(X)$ is evaluated on the fixed inverse branch. The defining identity is

$$\Phi(e^{w_0}) = \Phi(r_0) = X = e^Y,$$

or equivalently

$$X = \lambda\frac{e^{2w_0}}{H(w_0)}.$$

Thus

$$2w_0 = Y + \log H(w_0) - \log\lambda.$$

Since $X \to \infty$ and $\Phi(e^w) \to \infty$ on the chosen tail, we have $w_0 \to \infty$. Also, $H(w) = w + O_B(\log w)$, so $\log H(w_0) = o(w_0)$. The identity above gives $Y = 2w_0 + o(w_0)$, and hence

$$w_0 \sim \frac{Y}{2}.$$

Consequently,

$$\log H(w_0) = O_B(\log Y),$$

and the preceding identity improves to

$$w_0 = \frac{Y}{2} + O_B(\log Y).$$

Therefore

$$H(w_0) = \frac{Y}{2} + O_B(\log Y),$$

and so

$$\log H(w_0) = \log Y - \log 2 + o(1).$$

The fixed constant B enters only inside the $O_B(\log Y)$ term before this logarithm; after the logarithm it contributes no constant term. Substituting the last estimate into

$$2w_0 = Y + \log H(w_0) - \log \lambda$$

gives (23). Exponentiating also gives the explicit endpoint asymptotic

$$r_0 = \psi(X) = \sqrt{\frac{XY}{2\lambda}}\,(1 + o(1)).$$

For fixed Q, the stated consequence follows because

$$\frac{r_0}{X/Q} = Q\sqrt{\frac{Y}{2\lambda X}}\,(1 + o(1)) \to 0.$$

□

Lemma 8 (Square-term asymptotic and constant computation)**.** *As $X \to \infty$ through positive integers,*

$$X \int_X^M f(t)^2\,dt = X + \frac{\lambda\big(1 - 2B + 2\log(\lambda/2)\big)X}{Y} + o\left(\frac{X}{Y}\right).$$

Proof. This is where the $1/Y$ constant in the continuous majorant is determined.

As observed in Section 3, $M = \Phi(X) \asymp_B X^2/Y$, so $M > X$ for all sufficiently large X. For all sufficiently large X, we also have $X > T_*$ and $X \geq N_0$. Hence $\psi(X) \geq N_0$, and throughout the range $X \leq t \leq M$, every argument of ψ lies on the inverse $\psi : [T_*, \infty) \to [N_0, \infty)$ of $\Phi : [N_0, \infty) \to [T_*, \infty)$. This is the only branch used in all changes of variables below.

Use the change of variables

$$t = \Phi(e^w) = \lambda \frac{e^{2w}}{H(w)}.$$

Then

$$\psi(t) = e^w, \qquad f(t) = e^{-w}.$$

Also,

$$dt = \lambda e^{2w}\left(\frac{2}{H(w)} - \frac{H'(w)}{H(w)^2}\right) dw.$$

Therefore

$$f(t)^2\,dt = \lambda\left(\frac{2}{H(w)} - \frac{H'(w)}{H(w)^2}\right) dw. \tag{24}$$

The lower endpoint $t = X$ corresponds to $w = w_0 = \log \psi(X)$. The upper endpoint satisfies $M = \Phi(X) = \Phi(e^Y)$, so under $t = \Phi(e^w)$ it corresponds to $e^w = X$, hence to $w = Y$. By (23), $w_0 < Y$ for all sufficiently large X, so the interval $[w_0, Y]$ is ordered as written. Since

$$\left(\frac{1}{H(w)}\right)' = -\frac{H'(w)}{H(w)^2},$$

(24) gives

$$\int_X^M f(t)^2\,dt = \lambda\left[2\int_{w_0}^Y \frac{dw}{H(w)} + \frac{1}{H(Y)} - \frac{1}{H(w_0)}\right]. \tag{25}$$

The estimate (23) gives $w_0 \sim Y/2$. Hence the interval $[w_0, Y]$ has length $O_B(Y)$, and uniformly for $w \in [w_0, Y]$, we have $w \asymp Y$. Also,

$$H(w) = w\left(1 - \frac{\log w - B}{w}\right).$$

Since

$$\frac{\log w - B}{w} = O_B\left(\frac{\log Y}{Y}\right)$$

uniformly on $[w_0, Y]$, after increasing the lower threshold in X we may assume

$$\left|\frac{\log w - B}{w}\right| \le \frac{1}{2} \qquad (w \in [w_0, Y]).$$

The geometric expansion gives

$$\frac{1}{H(w)} = \frac{1}{w} + \frac{\log w - B}{w^2} + O_B\left(\frac{(\log Y)^2}{Y^3}\right). \tag{26}$$

The error term in (26), integrated over an interval of length $O_B(Y)$, contributes

$$O_B\left(\frac{(\log Y)^2}{Y^2}\right) = o\left(\frac{1}{Y}\right).$$

Write

$$w_0 = \frac{Y}{2} + r_X.$$

By (23),

$$r_X = \frac{1}{2}\log Y - \frac{1}{2}\log(2\lambda) + o(1). \tag{27}$$

This moving lower endpoint is the source of the $-2\log Y$ contribution that cancels the later $+2\log Y$ term in the square-term constant computation. Since $r_X = O_B(\log Y)$,

$$w_0 = \frac{Y}{2}\left(1 + \frac{2r_X}{Y}\right), \qquad \frac{r_X^2}{Y^2} = O\left(\frac{(\log Y)^2}{Y^2}\right) = o\left(\frac{1}{Y}\right).$$

Therefore Taylor expansion of $\log(1 + 2r_X/Y)$ gives

$$\log w_0 = \log\frac{Y}{2} + \frac{2r_X}{Y} + o\left(\frac{1}{Y}\right),$$

and hence

$$2\int_{w_0}^{Y}\frac{dw}{w} = 2\log 2 - \frac{4r_X}{Y} + o\left(\frac{1}{Y}\right).$$

Also,

$$\int \frac{\log w - B}{w^2}\,dw = -\frac{\log w + 1 - B}{w}.$$

Thus

$$2\int_{w_0}^{Y}\frac{\log w - B}{w^2}\,dw = 2\left[-\frac{\log Y + 1 - B}{Y} + \frac{\log w_0 + 1 - B}{w_0}\right].$$

Using $w_0 = Y/2 + r_X$ and $r_X = O_B(\log Y)$, the errors caused by r_X in the denominator and logarithm are

$$O\left(\frac{(\log Y)^2}{Y^2}\right) = o\left(\frac{1}{Y}\right).$$

Therefore

$$\frac{\log w_0 + 1 - B}{w_0} = \frac{2(\log Y - \log 2 + 1 - B)}{Y} + o\left(\frac{1}{Y}\right).$$

Hence

$$2\int_{w_0}^{Y}\frac{\log w - B}{w^2}\,dw = \frac{2\log Y - 4\log 2 + 2 - 2B}{Y} + o\left(\frac{1}{Y}\right).$$

Finally,

$$H(Y) = Y + O_B(\log Y), \qquad H(w_0) = \frac{Y}{2} + O(\log Y).$$

Thus

$$\frac{1}{H(Y)} = \frac{1}{Y} + O\left(\frac{\log Y}{Y^2}\right), \qquad \frac{1}{H(w_0)} = \frac{2}{Y} + O\left(\frac{\log Y}{Y^2}\right).$$

Hence

$$\frac{1}{H(Y)} - \frac{1}{H(w_0)} = -\frac{1}{Y} + o\left(\frac{1}{Y}\right). \tag{28}$$

Square-term estimate. Combining (25) through (28), the three numerators of the $1/Y$ contributions inside the square brackets, before substituting the asymptotic for r_X, are respectively

$$\begin{aligned} 2\int_{w_0}^{Y}\frac{dw}{w} &: -4r_X, \\ 2\int_{w_0}^{Y}\frac{\log w - B}{w^2}\,dw &: 2\log Y - 4\log 2 + 2 - 2B, \\ \frac{1}{H(Y)} - \frac{1}{H(w_0)} &: -1. \end{aligned}$$

Equivalently, (27) gives

$$-4r_X = -2\log Y + 2\log(2\lambda) + o(1),$$

which is the term cancelling the positive $2\log Y$ contribution below. Therefore, after using (27), the resulting coefficient is

$$\begin{aligned} &\big(-4r_X\big) + \big(2\log Y - 4\log 2 + 2 - 2B\big) - 1 \\ &= \big(-2\log Y + 2\log(2\lambda)\big) + \big(2\log Y - 4\log 2 + 2 - 2B\big) - 1 + o(1) \\ &= 1 - 2B + 2\log(2\lambda) - 4\log 2 + o(1) \\ &= 1 - 2B + 2\log(\lambda/2) + o(1). \end{aligned}$$

Thus the logarithmic terms cancel explicitly; the last line uses $2\log(2\lambda) - 4\log 2 = 2\log(\lambda/2)$. Also $2\lambda\log 2 = 1$ and $\lambda/2 = (4\log 2)^{-1}$. We obtain

$$\int_X^M f(t)^2\,dt = \lambda\left[2\log 2 + \frac{1 - 2B + 2\log(\lambda/2)}{Y}\right] + o\left(\frac{1}{Y}\right).$$

Since $2\lambda\log 2 = 1$, this becomes

$$\int_X^M f(t)^2\,dt = 1 + \frac{\lambda\big(1 - 2B + 2\log(\lambda/2)\big)}{Y} + o\left(\frac{1}{Y}\right).$$

Thus

$$X\int_X^M f(t)^2\,dt = X + \frac{\lambda\big(1 - 2B + 2\log(\lambda/2)\big)X}{Y} + o\left(\frac{X}{Y}\right). \tag{29}$$

This is the square-term asymptotic used in the final coefficient comparison.
Square-term constant computation. In the bracket in (25), the only $1/Y$-scale contributions are

$$\begin{aligned} \text{lower-endpoint displacement}: &\quad -4r_X, \\ \text{integral of } (\log w - B)/w^2: &\quad 2\log Y - 4\log 2 + 2 - 2B, \\ \text{endpoint denominator difference}: &\quad -1. \end{aligned}$$

Using $r_X = \frac{1}{2}\log Y - \frac{1}{2}\log(2\lambda) + o(1)$, these add to

$$1 - 2B + 2\log(\lambda/2) + o(1),$$

which is the coefficient in (29). No $1/Y$-scale term is left in the error: the integrated remainder in (26), the r_X-dependent denominator and logarithm errors, and the endpoint expansion errors are all $o(1/Y)$ before multiplication by X. □

6.2. The correction term

L^1 rescaling lemma

Lemma 9 (L^1 rescaling)**.** *Define*

$$g_X(v) := \sqrt{\frac{XY}{2\lambda}}\,f(Xv) \qquad (v > 0).$$

The value at $v = 0$ is irrelevant to all integrals below. For every fixed real $U > 0$,

$$g_X \to \big(v \mapsto v^{-1/2}\big) \qquad \text{in } L^1((0,U)). \tag{30}$$

Proof. Fix $0 < \varepsilon < \min\{U, 1\}$. The parameter ε is fixed before the limit $X \to \infty$ is taken; only after the limsup estimate is obtained do we let $\varepsilon \downarrow 0$. The convergence is uniform on every interval $[\varepsilon, U]$, by (5). On $(0, \varepsilon)$,

$$\int_0^\varepsilon g_X(v)\,dv = \sqrt{\frac{Y}{2\lambda X}}\int_0^{\varepsilon X} f(s)\,ds.$$

Choose a fixed $T_1 \geq T_*$ large enough for (6) and (8) to hold on $[T_1, \infty)$. The compact initial part contributes

$$\sqrt{\frac{Y}{2\lambda X}}\int_0^{T_1} f(s)\,ds = O_{A,B}\left(\sqrt{\frac{Y}{X}}\right) = o(1).$$

For each fixed $\varepsilon > 0$ and all sufficiently large X, say with $\varepsilon X \geq T_1$, the remaining part satisfies, by (6) and (8),

$$\int_{T_1}^{\varepsilon X} f(s)\,ds \ll_B \int_{T_1}^{\varepsilon X} \frac{ds}{\sqrt{s\log s}} \ll_B \sqrt{\frac{\varepsilon X}{\log(\varepsilon X)}}.$$

Consequently,

$$\int_0^{\varepsilon X} f(s)\,ds \ll_{A,B} 1 + \sqrt{\frac{\varepsilon X}{\log(\varepsilon X)}}.$$

For fixed $\varepsilon > 0$ and all sufficiently large X depending on this fixed ε,

$$\log(\varepsilon X) = Y + \log\varepsilon, \qquad \limsup_{X\to\infty} \sqrt{\frac{Y}{\log(\varepsilon X)}} = 1.$$

Thus, for fixed $0 < \varepsilon \leq \min\{U, 1\}$, taking $X \to \infty$ first with a post-limsup constant independent of both ε and U gives the small-v uniform-integrability estimate

$$\limsup_{X\to\infty} \int_0^{\varepsilon} g_X(v)\,dv \leq C_{A,B}\sqrt{\varepsilon}.$$

Together with

$$\int_0^{\varepsilon} v^{-1/2}\,dv = 2\sqrt{\varepsilon},$$

this yields

$$\limsup_{X\to\infty} \int_0^{\varepsilon} |g_X(v) - v^{-1/2}|\,dv \ll_{A,B} \sqrt{\varepsilon},$$

where the implicit constant is independent of $0 < \varepsilon \leq \min\{U, 1\}$. The threshold in X may depend on the fixed value of ε; only afterward do we let $\varepsilon \downarrow 0$.

Combining this with uniform convergence on $[\varepsilon, U]$, we have

$$\limsup_{X\to\infty} \|g_X - v^{-1/2}\|_{L^1(0,U)} \leq C_{A,B}\sqrt{\varepsilon} + \limsup_{X\to\infty} \|g_X - v^{-1/2}\|_{L^1(\varepsilon,U)}.$$

The second term is 0, and then letting $\varepsilon \downarrow 0$ proves (30). □

Truncated correction term

In this subsection the order of limits is fixed: first fix $U \geq 2$, then let $X \to \infty$, and only after the normalized estimates are obtained let $U \to \infty$. The parameter U never depends on X.

Fix a real number $U \geq 2$, and define

$$R_U(X) := \int_X^{UX} f(t) \int_{t-X}^{t} \big(f(s) - f(t)\big)\,ds\,dt.$$

Here U is fixed as $X \to \infty$. For fixed U, one has $UX < M$ for all sufficiently large X, because

$$\frac{M}{UX} \asymp_{B,U} \frac{X}{Y} \to \infty.$$

Let

$$h(u) := 2\left(1 - \sqrt{1 - \frac{1}{u}}\right) - \frac{1}{u} \qquad (u \geq 1).$$

With

$$t = X\alpha, \qquad s = X\beta,$$

we get

$$\frac{Y}{2\lambda X} R_U(X) = \int_1^U g_X(\alpha) \int_{\alpha-1}^{\alpha} \big(g_X(\beta) - g_X(\alpha)\big)\, d\beta\, d\alpha.$$

On $[1, U]$, $g_X(\alpha) \to \alpha^{-1/2}$ uniformly, again by (5), and the functions g_X are uniformly bounded there for all sufficiently large X. Also,

$$\beta^{-1/2} \in L^1(0, U).$$

Define

$$K(\varphi)(\alpha) := \int_{\alpha-1}^{\alpha} \varphi(\beta)\, d\beta.$$

Endpoint values are irrelevant for these Lebesgue integrals; in particular the integrable singularity of $\beta^{-1/2}$ at $\beta = 0$ causes no ambiguity at $\alpha = 1$. Then

$$\|K(\varphi)\|_{L^\infty(1,U)} \le \|\varphi\|_{L^1(0,U)}.$$

By (30),

$$K(g_X) \to K(\beta^{-1/2})$$

uniformly on $[1, U]$. Moreover, $K(\beta^{-1/2})(\alpha)$ is finite and continuous on $[1, U]$. The $L^1(0, U)$ convergence also gives $\|g_X\|_{L^1(0,U)} = O_{A,B,U}(1)$, so $K(g_X)$ is uniformly bounded on $[1, U]$; and g_X is uniformly bounded on $[1, U]$ for all sufficiently large X. Writing $g(\alpha) = \alpha^{-1/2}$, for $\alpha \in [1, U]$ we have

$$\begin{aligned} &|g_X(\alpha)K(g_X)(\alpha) - g(\alpha)K(g)(\alpha)| \\ &\le \|g_X - g\|_{L^\infty(1,U)} \|K(g_X)\|_{L^\infty(1,U)} \\ &\quad + \|g\|_{L^\infty(1,U)} \|K(g_X) - K(g)\|_{L^\infty(1,U)}, \end{aligned}$$

and

$$|g_X(\alpha)^2 - g(\alpha)^2| \le (\|g_X\|_{L^\infty(1,U)} + \|g\|_{L^\infty(1,U)}) \|g_X - g\|_{L^\infty(1,U)}.$$

Both right-hand sides tend to 0. Hence

$$g_X(\alpha)K(g_X)(\alpha) - g_X(\alpha)^2 \to \alpha^{-1/2} K(\beta^{-1/2})(\alpha) - \alpha^{-1}$$

uniformly on $[1, U]$, and therefore in $L^1(1, U)$. Consequently,

$$\frac{Y}{2\lambda X} R_U(X) \to \int_1^U \left[\alpha^{-1/2} \int_{\alpha-1}^{\alpha} \beta^{-1/2}\, d\beta - \frac{1}{\alpha} \right] d\alpha.$$

Since

$$\alpha^{-1/2} \int_{\alpha-1}^{\alpha} \beta^{-1/2}\, d\beta = 2\left(1 - \sqrt{1 - \frac{1}{\alpha}}\right),$$

we get

$$R_U(X) = \frac{2\lambda X}{Y} \int_1^U \left[2\left(1 - \sqrt{1 - \frac{1}{\alpha}}\right) - \frac{1}{\alpha} \right] d\alpha + o_{A,B,U}\left(\frac{X}{Y}\right). \tag{31}$$

The limiting integrand is $h(\alpha)$, and

$$h(\alpha) = \frac{1}{4\alpha^2} + O\left(\frac{1}{\alpha^3}\right) \qquad (\alpha \to \infty).$$

At $\alpha = 1$ it equals 1, and it is continuous on $[1, 2]$. Hence the corresponding improper integral converges absolutely.

Correction-tail estimate with truncation-independent constant

Lemma 10 (Correction-tail estimate)**.** *For every fixed real $U \geq 2$,*

$$0 \leq R(X) - R_U(X) \ll_B \frac{X}{UY}, \tag{32}$$

where the implicit constant is independent of U, while the lower threshold for X may depend on the fixed value of U. Consequently,

$$\lim_{U\to\infty} \limsup_{X\to\infty} \frac{Y}{2\lambda X}\big(R(X) - R_U(X)\big) = 0.$$

Proof. Take a fixed real number $U \geq 2$. For fixed U, all sufficiently large X satisfy $UX < M$. If $t \geq UX$ and $s \in [t - X, t]$, then

$$s \in [t/2, t].$$

For sufficiently large X, this interval lies in the smooth tail (T_*, ∞). The lower threshold for X in this paragraph may depend on the fixed value of U, but the constant in (12) with $c = 1/2$ depends only on B. Applying (12) with $c = 1/2$, we get

$$0 \leq f(s) - f(t) \ll_B \frac{f(t)}{t}(t - s).$$

Therefore

$$\int_{t-X}^{t} \big(f(s) - f(t)\big)\, ds \ll_B \frac{f(t)X^2}{t}.$$

Hence

$$0 \leq R(X) - R_U(X) \ll_B X^2 \int_{UX}^{M} \frac{f(t)^2}{t}\, dt.$$

By (5),

$$f(t)^2 \ll_B \frac{1}{t \log t},$$

and for $t \geq UX$ with $U \geq 2$ we have $\log t \geq \log(UX) \geq Y$. Thus, enlarging the upper limit to infinity, uniformly in $U \geq 2$ in the constant, and with only the lower threshold in X allowed to depend on the fixed U,

$$R(X) - R_U(X) \ll_B X^2 \int_{UX}^{\infty} \frac{dt}{t^2 \log t}.$$

Moreover,

$$\int_{UX}^{\infty} \frac{dt}{t^2 \log t} \leq \frac{1}{Y} \int_{UX}^{\infty} \frac{dt}{t^2} = \frac{1}{UXY}.$$

Therefore

$$R(X) - R_U(X) \ll_B \frac{X}{UY}.$$

This proves (32). Equivalently, for every fixed $U \geq 2$,

$$\limsup_{X\to\infty} \frac{Y}{2\lambda X}\big(R(X) - R_U(X)\big) \leq \frac{C_B}{U},$$

where the constant is independent of U. Hence

$$\lim_{U\to\infty} \limsup_{X\to\infty} \frac{Y}{2\lambda X}\big(R(X) - R_U(X)\big) = 0.$$

Here the threshold for X may depend on the fixed value of U, while the implicit constant in (32) does not. □

From (31) and the preceding tail estimate, for every fixed $U \geq 2$,

$$\limsup_{X\to\infty} \left| \frac{Y}{2\lambda X} R(X) - \int_1^\infty h(u)\, du \right| \leq \frac{C_B}{U} + \int_U^\infty |h(u)|\, du.$$

Here we used $R(X) - R_U(X) \geq 0$ and the normalized tail bound (32); no sign information on h is needed because $\int_U^\infty |h(u)|\, du \to 0$. Since $h(u) = O(u^{-2})$, the right-hand side tends to 0 as $U \to \infty$. Therefore

$$R(X) = \frac{2\lambda X}{Y} \int_1^\infty \left[2\left(1 - \sqrt{1 - \frac{1}{u}}\right) - \frac{1}{u} \right] du + o\left(\frac{X}{Y}\right).$$

With

$$z = \sqrt{1 - \frac{1}{u}},$$

one has

$$u = \frac{1}{1 - z^2}, \qquad du = \frac{2z}{(1 - z^2)^2}\, dz.$$

Moreover,

$$h(u)\, du = (1 - z)^2 \frac{2z}{(1 - z^2)^2}\, dz = \frac{2z}{(1 + z)^2}\, dz.$$

The integral becomes

$$\int_0^1 \frac{2z}{(1 + z)^2}\, dz.$$

Since

$$\int \frac{2z}{(1 + z)^2}\, dz = 2\log(1 + z) + \frac{2}{1 + z},$$

evaluation at 0 and 1 gives

$$\int_0^1 \frac{2z}{(1 + z)^2}\, dz = 2\log 2 - 1.$$

Therefore

$$R(X) = \frac{2\lambda(2\log 2 - 1)X}{Y} + o\left(\frac{X}{Y}\right). \tag{33}$$

6.3. Assembling $J_B(X)$

The continuous-majorant coefficient now has three sources only: the small-top square term (21), the square term (29), and the correction term (33). Keeping these sources separate, from (21), (22), (29), and (33),

$$J_B(X) = \frac{4\lambda X}{Y} + X + \frac{\lambda\big(1 - 2B + 2\log(\lambda/2)\big)X}{Y} + \frac{2\lambda(2\log 2 - 1)X}{Y} + o\left(\frac{X}{Y}\right).$$

Thus

$$J_B(X) = X + \frac{\lambda X}{Y}\left[4 + 1 - 2B + 2\log(\lambda/2) + 2(2\log 2 - 1)\right] + o\left(\frac{X}{Y}\right).$$

The bracket equals

$$3 - 2B + 4\log 2 + 2\log(\lambda/2).$$

Since

$$\lambda = \frac{1}{2\log 2}, \qquad \frac{\lambda}{2} = \frac{1}{4\log 2},$$

we have

$$2\log(\lambda/2) = -4\log 2 - 2\log(\log 2),$$

and therefore

$$4\log 2 + 2\log(\lambda/2) = -2\log(\log 2).$$

Therefore

$$J_B(X) = X + \frac{\lambda\big(3 - 2B - 2\log(\log 2)\big)X}{Y} + o\left(\frac{X}{Y}\right),$$

which proves (20).

7. The floor saving

Define

$$E(X) := \int_X^M f(t)\{G_X(t)\}\,dt,$$

where

$$\{x\} := x - \lfloor x \rfloor.$$

Since $0 \le \{G_X(t)\} < 1$, we have the trivial bounds

$$0 \le E(X) \le \int_X^M f(t)\,dt.$$

Then

$$\int_X^M f(t)\lfloor G_X(t)\rfloor\,dt = \int_X^M f(t)G_X(t)\,dt - E(X). \tag{34}$$

The floor saving is the following lemma.

Lemma 11 (Floor saving)**.** *With the definitions above,*

$$E(X) = 2\lambda(1-\gamma)\frac{X}{\log X} + o\left(\frac{X}{\log X}\right). \tag{35}$$

Proof. The proof has four separate steps: a fixed-Q change of variables, a uniform approximation for $G_X(\Phi(X/q))$, removal of the fractional-part discontinuities on the fixed interval $1 \le q \le Q$, and a large-q tail bound with a normalized constant independent of Q. The relevant discontinuities are those of $q \mapsto \{q\}$ at the integers; this is why the fixed-Q convergence below first removes small neighborhoods of the finitely many integers in $[1, Q]$.

Put

$$q(t) := Xf(t) = \frac{X}{\psi(t)}.$$

All occurrences of ψ in this part are on the inverse branch fixed above; after fixing Q, the lower threshold in X is enlarged whenever necessary so that all displayed arguments of Φ and ψ lie on this branch. Since ψ is increasing, $q(t)$ is decreasing in t; larger q corresponds to smaller t in the large-top interval. Equivalently, if $r = \psi(t)$, then

$$q = \frac{X}{r}, \qquad t = \Phi(r) = \Phi(X/q).$$

Fix a real number $Q > 1$. In this part, Q is fixed before $X \to \infty$. All lower bounds on X may depend on Q, and all $o_{A,B,Q}$-terms may depend on A, B, Q and the fixed threshold data, but not on X. Only after the limiting sandwich is obtained do we let $Q \to \infty$.

For fixed $Q > 1$, we have

$$\Phi(X/Q) \asymp_{B,Q} \frac{X^2}{Y} > X$$

for all sufficiently large X. Since $Q > 1$ and Φ is increasing on the chosen branch,

$$X < \Phi(X/Q) < M = \Phi(X)$$

for all sufficiently large X. Thus $G_X(\Phi(X/q))$ is defined for every $q \in [1, Q]$ once X is large enough.

Step 1: fixed-Q q-change of variables

Lemma 12 (Fixed-Q q-change of variables)**.** *Here $Q > 1$ is fixed before $X \to \infty$; no assertion is made with $Q = Q(X)$. For every bounded measurable function $\Theta_X : [1, Q] \to \mathbb{R}$ satisfying $|\Theta_X| \le 1$,*

$$\int_{\Phi(X/Q)}^{M} f(t)\Theta_X(q(t))\,dt = \frac{2\lambda X}{Y} \int_1^Q \frac{\Theta_X(q)}{q^2}\,dq + o_{A,B,Q}\left(\frac{X}{Y}\right), \tag{36}$$

uniformly over all such measurable choices of Θ_X.

Proof. The uniformity is over the bounded measurable test functions for this fixed Q; it is not a statement with $Q = Q(X)$. The function Θ_X is allowed to depend on X, and no continuity or regularity of Θ_X is used; the error is controlled below by a supremum before integration. For fixed $Q > 1$, assume X is so large that $X/Q \ge N_0$. On the interval

$$\Phi(X/Q) \le t \le M = \Phi(X),$$

the monotonicity of Φ on $[N_0, \infty)$ and the inverse relation between Φ and ψ give

$$X/Q \le \psi(t) \le X,$$

and therefore

$$1 \le q(t) \le Q.$$

The monotone C^1 change-of-variables theorem applies to these bounded measurable integrands. Changing variables $r = \psi(t)$, and then $q = X/r$, gives the endpoint correspondence

$$t = \Phi(X/Q) \iff q = Q, \qquad t = M = \Phi(X) \iff q = 1.$$

Hence

$$\begin{aligned}\int_{\Phi(X/Q)}^{M} f(t)\Theta_X(q(t))\,dt &= \int_{X/Q}^{X} \frac{1}{r}\Theta_X(X/r)\Phi'(r)\,dr \\ &= \int_{Q}^{1} \Theta_X(q)\frac{\Phi'(X/q)}{X/q}\left(-\frac{X}{q^2}\right)dq \\ &= \int_{1}^{Q} \Theta_X(q)\frac{\Phi'(X/q)}{X/q}\frac{X}{q^2}\,dq.\end{aligned}$$

We keep the product in the final integrand in this form because the first factor is uniformly approximated by $2\lambda/Y + O_{B,Q}(\log Y/Y^2)$. With Q fixed, for $1 \le q \le Q$, put

$$w = \log(X/q) = Y - \log q.$$

Then uniformly for $1 \le q \le Q$,

$$H(w) = w - \log w + B = Y - \log Y + B - \log q + O_Q\left(\frac{1}{Y}\right) = Y + O_{B,Q}(\log Y),$$

and

$$H'(w) = 1 - \frac{1}{w} = 1 + O_Q\left(\frac{1}{Y}\right).$$

Therefore, uniformly for $1 \le q \le Q$,

$$\frac{\Phi'(X/q)}{X/q} = \lambda\frac{2H(w) - H'(w)}{H(w)^2} = \frac{2\lambda}{Y} + O_{B,Q}\left(\frac{\log Y}{Y^2}\right).$$

In particular,

$$\sup_{1\le q\le Q}\left|\frac{\Phi'(X/q)}{X/q} - \frac{2\lambda}{Y}\right| = o_{A,B,Q}\left(\frac{1}{Y}\right).$$

Thus the total error in replacing $\Phi'(X/q)/(X/q)$ by $2\lambda/Y$ is bounded by

$$X\sup_{1\le q\le Q}\left|\frac{\Phi'(X/q)}{X/q} - \frac{2\lambda}{Y}\right|\int_1^Q \frac{dq}{q^2} = o_{A,B,Q}\left(\frac{X}{Y}\right),$$

uniformly over all measurable Θ_X with $|\Theta_X| \le 1$. The constant is uniform because Q is fixed before the limit $X \to \infty$; no uniformity in growing Q is asserted here. This proves (36). □

Lemma 13 (Uniform approximation to q)**.** *Uniformly for* $1 \le q \le Q$*,*

$$G_X(\Phi(X/q)) = q + O_{B,Q}\left(\frac{Y}{X}\right). \tag{37}$$

Proof. Put

$$t = \Phi(X/q), \qquad 1 \le q \le Q.$$

Then, since $H(Y - \log q) = Y + O_{B,Q}(\log Y)$,

$$t = \lambda \frac{(X/q)^2}{H(Y - \log q)} \asymp_{B,Q} \frac{X^2}{q^2 Y}, \qquad \frac{X}{t} = O_B\left(\frac{q^2 Y}{X}\right) = O_{B,Q}\left(\frac{Y}{X}\right),$$

and also

$$\frac{qX}{t} = O_B\left(\frac{q^3 Y}{X}\right) = O_{B,Q}\left(\frac{Y}{X}\right).$$

In particular $t \to \infty$ uniformly for $1 \le q \le Q$, so the threshold in (12) with $c = 1/2$ is exceeded uniformly for fixed Q. For $s \in [t - X, t]$, we have $s \in [t/2, t]$ for all sufficiently large X. By (12),

$$f(s) = f(t) + O_B\left(\frac{f(t)}{t} X\right)$$

uniformly in $q \in [1, Q]$. Therefore

$$G_X(t) = \int_{t-X}^{t} f(s)\, ds = X f(t) + O_B\left(\frac{f(t) X^2}{t}\right).$$

Since

$$\psi(\Phi(X/q)) = X/q, \qquad f(\Phi(X/q)) = \frac{q}{X},$$

we obtain, using the preceding bound on qX/t,

$$G_X(t) = q + O_B\left(\frac{qX}{t}\right) = q + O_B\left(\frac{q^3 Y}{X}\right) = q + O_{B,Q}\left(\frac{Y}{X}\right),$$

which proves (37). □

Step 2: fractional-part convergence away from discontinuities

Lemma 14 (Fractional-part convergence on a fixed q-range)**.** *For every fixed real $Q > 1$,*

$$\int_{\Phi(X/Q)}^{M} f(t)\{G_X(t)\}\, dt = \frac{2\lambda X}{Y} \int_1^Q \frac{\{q\}}{q^2}\, dq + o_{A,B,Q}\left(\frac{X}{Y}\right). \tag{38}$$

Proof. The following compactness argument is the discontinuity-removal step for the fractional-part map. Since $q \mapsto \Phi(X/q)$ is continuous on $[1, Q]$ and f is locally integrable, the primitive

$$P(t) := \int_0^t f(s)\, ds$$

is absolutely continuous on compact intervals, and hence continuous. For $t \ge X$,

$$G_X(t) = P(t) - P(t - X),$$

so G_X is continuous on $[X, \infty)$. Hence the function

$$\Theta_X(q) := \{G_X(\Phi(X/q))\}$$

is measurable. Since (36) is uniform over all measurable $|\Theta_X| \le 1$, it may be applied to this X-dependent choice.

The only obstruction to replacing $\{G_X(\Phi(X/q))\}$ by $\{q\}$ is crossing an integer; the next paragraph removes small neighborhoods of the finitely many relevant integers. For fixed $Q > 1$ and $\eta > 0$, choose $\mathcal{U}_\eta \subset [1, Q]$ to be a finite union of small relatively open intervals around the integers in $[1, Q]$, using one-sided intervals at endpoint integers if present, such that

$$\int_{\mathcal{U}_\eta} \frac{dq}{q^2} < \eta.$$

This is possible because there are only finitely many integers in $[1, Q]$. Endpoint behavior is irrelevant for the Lebesgue integral; including endpoint integers only makes the uniform convergence statement on the complement unambiguous. Let

$$S_\eta := [1, Q] \setminus \mathcal{U}_\eta.$$

If $S_\eta = \varnothing$, the convergence claim below is vacuous. Otherwise, S_η is compact and has positive distance from the finite set $\mathbb{Z} \cap [0, Q + 1]$; all other integers are farther away from $[1, Q]$. Call this distance $\rho_\eta > 0$. Thus

$$\operatorname{dist}(q, \mathbb{Z}) \geq \rho_\eta \qquad (q \in S_\eta).$$

By the uniform convergence (37), for large X,

$$|G_X(\Phi(X/q)) - q| < \frac{\rho_\eta}{2}$$

uniformly for $q \in S_\eta$. Therefore no integer is crossed: $G_X(\Phi(X/q))$ and q lie in the same connected component of $\mathbb{R} \setminus \mathbb{Z}$. On the relevant component away from the removed neighborhoods, the fractional-part map is Lipschitz with constant 1, so

$$|\{G_X(\Phi(X/q))\} - \{q\}| \leq |G_X(\Phi(X/q)) - q|.$$

Hence

$$\{G_X(\Phi(X/q))\} \to \{q\}$$

uniformly for $q \in S_\eta$.

Applying (36) to

$$\Theta_X(q) := \{G_X(\Phi(X/q))\},$$

and using that, for $t \in [\Phi(X/Q), M]$,

$$\Phi(X/q(t)) = \Phi(\psi(t)) = t,$$

we get

$$\int_{\Phi(X/Q)}^{M} f(t)\{G_X(t)\}\, dt = \frac{2\lambda X}{Y} \int_1^Q \frac{\{G_X(\Phi(X/q))\}}{q^2}\, dq + o_{A,B,Q}\left(\frac{X}{Y}\right).$$

On S_η, the integrand converges uniformly to $\{q\}/q^2$. On $\mathcal{U}_\eta$, both fractional parts lie in $[0, 1)$, so the total possible discrepancy is at most

$$\int_{\mathcal{U}_\eta} \frac{dq}{q^2} < \eta.$$

Therefore

$$\limsup_{X \to \infty} \left| \int_1^Q \frac{\{G_X(\Phi(X/q))\} - \{q\}}{q^2}\, dq \right| \leq \eta.$$

Letting $\eta \downarrow 0$, we get (38). □

Step 3: large-q floor-tail estimate with Q-independent constant

Lemma 15 (Fixed-Q large-q floor-tail estimate)**.** *For every fixed real $Q > 1$, fixed independently of X, all sufficiently large X satisfy*

$$\int_X^{\Phi(X/Q)} f(t)\{G_X(t)\}\,dt \le C_B \frac{X}{QY}, \tag{39}$$

where C_B is independent of Q, while the lower threshold for X may depend on the fixed value of Q.

Proof. It remains to control the range

$$X \le t \le \Phi(X/Q),$$

which is the large-q tail because $q(t) = X/\psi(t)$ decreases as t increases. Splitting $E(X)$ at the common endpoint $t = \Phi(X/Q)$ changes no integral, since that endpoint has measure zero. Only a normalized bound tending to zero as $Q \to \infty$ is needed here; no asymptotic for this range is required. By the lower-endpoint expansion (23), writing $r_0 = \psi(X)$,

$$\log r_0 = \frac{1}{2}Y + O_B(\log Y),$$

and hence $X/r_0 \to \infty$. Since ψ is increasing, $q(t) = X/\psi(t)$ is decreasing, with endpoint correspondence

$$t = X \iff q = \frac{X}{r_0}, \qquad t = \Phi(X/Q) \iff q = Q.$$

Thus this interval corresponds to $q \ge Q$ for all sufficiently large X. Endpoints do not affect the integral. For each fixed $Q > 1$, this interval is nonempty for all sufficiently large X, because

$$\Phi(X/Q) \asymp_{B,Q} \frac{X^2}{Y} \gg X.$$

Since

$$0 \le \{G_X(t)\} < 1,$$

we have

$$0 \le \int_X^{\Phi(X/Q)} f(t)\{G_X(t)\}\,dt \le \int_X^{\Phi(X/Q)} f(t)\,dt.$$

For each fixed $Q > 1$, the lower-endpoint lemma gives $r_0 = \psi(X) = o(X/Q)$, so all sufficiently large X satisfy

$$r_0 < X/Q.$$

This strict inequality fixes the orientation of the change of variables below. Here $X > T_*$ for all sufficiently large X, so $r_0 = \psi(X)$ is evaluated on the same inverse branch.

For fixed Q, all sufficiently large X also satisfy $X/Q \ge N_0$, so both endpoints are on the monotone branch of Φ. Changing variables $t = \Phi(r)$, the last integral becomes

$$\int_{r_0}^{X/Q} \frac{\Phi'(r)}{r}\,dr.$$

Now, for all sufficiently large r,

$$H(\log r) \asymp_B \log r, \qquad 2H(\log r) - H'(\log r) \asymp_B \log r.$$

Thus

$$\frac{\Phi'(r)}{r} = \lambda \frac{2H(\log r) - H'(\log r)}{H(\log r)^2} \ll_B \frac{1}{\log r}$$

uniformly for all $r \geq r_B$, with a constant depending only on B, after choosing a sufficiently large threshold r_B. Increase the lower threshold in X so that $r_0 \geq r_B$ and $X/Q \geq N_0$. Since

$$\log r_0 = \frac{1}{2}Y + O_B(\log Y),$$

we have $\log r_0 \geq Y/3$ for all sufficiently large X. Throughout this range, therefore,

$$\log r \geq \log r_0 \geq \frac{Y}{3}.$$

Consequently, using the interval orientation $r_0 < X/Q$ just established,

$$0 \leq \frac{X}{Q} - r_0 \leq \frac{X}{Q}.$$

Therefore

$$\int_{r_0}^{X/Q} \frac{\Phi'(r)}{r}\, dr \ll_B \int_{r_0}^{X/Q} \frac{dr}{Y} \leq C_B \frac{X}{QY}.$$

Therefore (39) follows. No sharper asymptotic is needed for this tail; the Q-uniform normalized $O(1/Q)$ bound is enough after $Q \to \infty$. Equivalently,

$$\frac{Y}{2\lambda X} \int_X^{\Phi(X/Q)} f(t)\{G_X(t)\}\, dt \ll_B \frac{1}{Q}. \tag{40}$$

Using the normalized form (40), the estimate also gives, in limsup form,

$$\limsup_{X\to\infty} \frac{Y}{2\lambda X} \int_X^{\Phi(X/Q)} f(t)\{G_X(t)\}\, dt \leq \frac{C_B}{Q},$$

with C_B independent of Q. Hence

$$\lim_{Q\to\infty} \limsup_{X\to\infty} \frac{Y}{2\lambda X} \int_X^{\Phi(X/Q)} f(t)\{G_X(t)\}\, dt = 0.$$

□

Combining (38) and (39), and keeping the order of limits $X \to \infty$ first with Q fixed and then $Q \to \infty$, gives the desired sandwich. The implicit constant in the normalized $O(1/Q)$ tail bound is independent of Q; only the lower threshold for X may depend on the fixed value of Q. This independence is the only property of the large-q tail used in the final sandwich. Let

$$\mathcal{I}_Q := \int_1^Q \frac{\{q\}}{q^2}\, dq, \qquad \mathcal{I} := \int_1^\infty \frac{\{q\}}{q^2}\, dq.$$

The integral defining $\mathcal{I}$ converges because

$$0 \leq \frac{\{q\}}{q^2} \leq \frac{1}{q^2}.$$

For fixed $Q > 1$, write

$$E_{\le Q}(X) := \int_{\Phi(X/Q)}^{M} f(t)\{G_X(t)\}\,dt, \qquad E_{>Q}(X) := \int_{X}^{\Phi(X/Q)} f(t)\{G_X(t)\}\,dt.$$

The subscripts refer to the corresponding q-range, not to the orientation of the t-intervals. Then $E(X) = E_{\le Q}(X) + E_{>Q}(X)$ up to the common endpoint, which has measure zero, and $E_{>Q}(X) \ge 0$. By (38),

$$\frac{Y}{2\lambda X} E_{\le Q}(X) \to \mathcal{I}_Q,$$

while the normalized tail estimate above, with a Q-independent constant, gives

$$0 \le \limsup_{X\to\infty} \frac{Y}{2\lambda X} E_{>Q}(X) \le \frac{C_B}{Q}.$$

In unnormalized form, the compact summary is

$$E_{\le Q}(X) = \frac{2\lambda X}{Y}\mathcal{I}_Q + o_{A,B,Q}\left(\frac{X}{Y}\right), \qquad 0 \le E_{>Q}(X) \ll_B \frac{X}{QY}$$

with $X \to \infty$ at fixed Q, and only then $Q \to \infty$. The order of limits in the next displays is

$$X \to \infty \quad \text{with } Q \text{ fixed, and then} \quad Q \to \infty.$$

For every fixed $Q > 1$, the preceding estimates imply

$$\liminf_{X\to\infty} \frac{YE(X)}{2\lambda X} \ge \mathcal{I}_Q, \qquad \limsup_{X\to\infty} \frac{YE(X)}{2\lambda X} \le \mathcal{I}_Q + \frac{C_B}{Q},$$

with C_B enlarged if necessary. Equivalently,

$$\mathcal{I}_Q \le \liminf_{X\to\infty} \frac{YE(X)}{2\lambda X} \le \limsup_{X\to\infty} \frac{YE(X)}{2\lambda X} \le \mathcal{I}_Q + \frac{C_B}{Q}.$$

Letting $Q \to \infty$, using $\mathcal{I}_Q \uparrow \mathcal{I}$ by monotone convergence and $C_B/Q \to 0$, gives the normalized convergence

$$\frac{YE(X)}{2\lambda X} \to \mathcal{I}.$$

Equivalently,

$$E(X) = \frac{2\lambda X}{Y}\int_1^\infty \frac{\{q\}}{q^2}\,dq + o\left(\frac{X}{Y}\right). \tag{41}$$

It remains to compute the integral. Endpoint values of the fractional-part function are immaterial here, since they affect only a measure-zero set. By nonnegativity, monotone convergence permits the split into unit intervals:

$$\int_1^\infty \frac{\{q\}}{q^2}\,dq = \sum_{m=1}^{\infty} \int_m^{m+1} \frac{q-m}{q^2}\,dq.$$

For each $m \ge 1$,

$$\int_m^{m+1} \frac{q-m}{q^2}\,dq = \log\frac{m+1}{m} - \frac{1}{m+1}.$$

Therefore

$$\sum_{m=1}^{K}\left(\log\frac{m+1}{m}-\frac{1}{m+1}\right)=\log(K+1)-\sum_{m=1}^{K}\frac{1}{m+1}.$$

Since

$$\sum_{m=1}^{K}\frac{1}{m+1}=\sum_{j=1}^{K+1}\frac{1}{j}-1,$$

we get

$$\sum_{m=1}^{K}\left(\log\frac{m+1}{m}-\frac{1}{m+1}\right)=\log(K+1)-\sum_{j=1}^{K+1}\frac{1}{j}+1.$$

Letting $K\to\infty$, and using the definition of γ,

$$\int_1^\infty\frac{\{q\}}{q^2}\,dq=1-\gamma.$$

This value is positive because the integrand is nonnegative and positive on a set of positive measure; numerically, $1-\gamma=0.42278\ldots$. Substituting this into (41) proves the floor-saving lemma, namely (35). □

8. Final contradiction

We have proved the two asymptotic evaluations

$$J_B(X)=X+\frac{\lambda\big(3-2B-2\log(\log 2)\big)X}{Y}+o\left(\frac{X}{Y}\right),$$

and

$$E(X)=2\lambda(1-\gamma)\frac{X}{Y}+o\left(\frac{X}{Y}\right).$$

Recall that X in this final comparison runs through sufficiently large positive integers. The final comparison uses (19), (34), (20), and (35); the continuous asymptotic (20) is assembled from (21), (29), and (33). From (19),

$$X\le\frac{1}{2}\left(\int_0^X f(t)\,dt\right)^2+\int_X^M f(t)\lfloor G_X(t)\rfloor\,dt+o\left(\frac{X}{Y}\right).$$

Using (34) and the definition of $J_B(X)$, this becomes

$$X\le J_B(X)-E(X)+o\left(\frac{X}{Y}\right).$$

By (20) and (35),

$$\begin{aligned}X\le X&+\frac{\lambda\big(3-2B-2\log(\log 2)\big)X}{Y}\\&-\frac{2\lambda(1-\gamma)X}{Y}+o\left(\frac{X}{Y}\right).\end{aligned}$$

The coefficient of X/Y is

$$\lambda\big(3-2B-2\log(\log 2)\big)-2\lambda(1-\gamma)=-2\lambda(B-B_1).$$

Let

$$\delta := B - B_1 > 0.$$

Hence the preceding inequality gives

$$0 \le -\frac{2\lambda\delta X}{Y} + o\left(\frac{X}{Y}\right).$$

After division by X/Y along integer $X \to \infty$, write the error term as $\eta_X = o(1)$. Thus

$$0 \le -2\lambda\delta + \eta_X.$$

For all sufficiently large integer X, $|\eta_X| \le \lambda\delta$, and hence

$$0 \le -2\lambda\delta + \eta_X \le -\lambda\delta < 0,$$

which is impossible.

Thus the assumed eventual upper bound (1) cannot hold. Given an integer N, choose an integer $N' \ge \max\{N, 2\}$. If every $n \ge N'$ satisfied the non-strict upper bound, then the forbidden eventual upper bound would hold from N' onward. Therefore there exists $n \ge N'$ such that

$$u(n) > \lambda \frac{n^2}{\log n - \log\log n + B}.$$

This proves the main assertion.

Fixed-constant limsup consequence

Fix a real constant C and fix one value $B^* > B_1$. Apply the main assertion just proved with this value of B^*. Then there is an unbounded sequence n_j such that

$$u(n_j) > \lambda \frac{n_j^2}{\log n_j - \log\log n_j + B^*}.$$

This uses no limiting process $B^* \downarrow B_1$ inside the contradiction argument and does not substitute $B = B_1$ in the pointwise inequality. Discarding finitely many terms if necessary, both quantities

$$\log n_j - \log\log n_j + C$$

and

$$\log n_j - \log\log n_j + B^*$$

are positive along this sequence. Therefore the following division and multiplication preserve the inequality sign, and along this unbounded sequence,

$$\frac{u(n_j)(\log n_j - \log\log n_j + C)}{n_j^2} > \lambda \frac{\log n_j - \log\log n_j + C}{\log n_j - \log\log n_j + B^*}.$$

The fixed additive difference $B^* - C$ is negligible compared with $\log n_j$; explicitly,

$$\frac{\log n_j - \log\log n_j + C}{\log n_j - \log\log n_j + B^*} = 1 - \frac{B^* - C}{\log n_j - \log\log n_j + B^*} \to 1.$$

The right-hand side tends to λ. Since the left-hand side is bounded below along this sequence by quantities tending to λ, the full limsup is at least λ. Hence, for every fixed real C,

$$\limsup_{n\to\infty} \frac{u(n)(\log n - \log\log n + C)}{n^2} \geq \lambda.$$

Since

$$\lambda = \frac{1}{2\log 2},$$

this is the fixed-constant limsup assertion in the theorem. Taking $C = B_1$ gives the endpoint limsup version there. Since

$$\frac{\log n}{\log n - \log\log n} \to 1,$$

the conclusion with $C = 0$ also gives

$$\limsup_{n\to\infty} \frac{u(n)\log n}{n^2} \geq \frac{1}{2\log 2}.$$

□

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